Numbers

Real

Imaginary

Fractions

Integers $(-\infty$ to $\infty)$ → Can be represented in a bumber line

Positive (Natural No.s) 1 to ∞

Zero 0

Negative -1 to $-\infty$

Pos Int + zero → whole No.s
(Nat No.)

Even
(2k)

odd
(2k±1)

Prime
exactly 2 factors

Composite
more than 2 factors

Smallest = 2 (Only Even Prime No.)

Prime No. $\xrightarrow{\text{always}}$ $6K\pm1$

$6K\pm1$ $\xrightarrow{\text{May or Not Be True}}$ Prime No.

13

$6\times2+1$

$17 = 6\times3-1$

$25 = 6\times4+1$

$1 \times 3 = 3 \quad (O \times O = O)$

$2 \times 3 = 6 \quad (O \times E = E)$

$4 \times 2 = 8 \quad [E \times E = E)$

$2 \times (\underline{1 \times 3 \times 5 \times 7}) =$

Even Odd = Even

$P_1 \times P_2 \times P_3 \times P_4 \text{ --- --- --- } \times P_{95} = 302080$

$P_1 = ?$

$P_1 = 2$

No.s

Rational

(In Number line)

$$\boxed{\frac{p}{q},\ q \neq 0}$$

$\boxed{\text{Irrational}}$

3.14 ✓ Rat

$\frac{1}{3} = 0.333\ldots$

$= 0.\overline{3}$

$\boxed{0.\overline{3}}$ = Rational

$\frac{22}{7}$ ✓ Rat

π ✗ Irrat

After Decimal, Non Terminating + Repeating = Rational

3.142873 - - - - -

$0.\bar{3} = 0.333$

Non Terminating & Repeating to fractions.

$0.\bar{3} = \frac{3}{9} = \frac{1}{3}$

$0.\overline{32} = \frac{32}{99}$

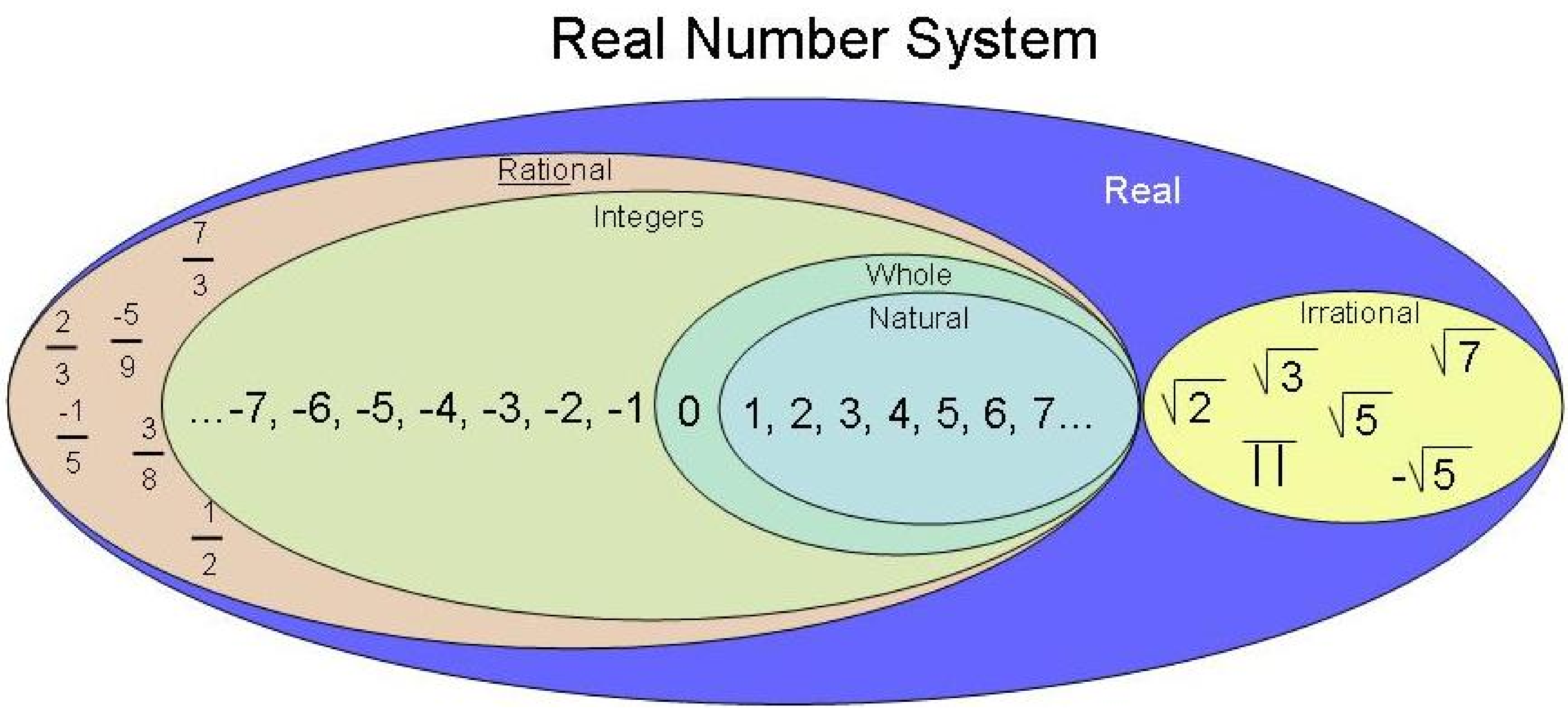
$2.\bar{3} = 2 + \frac{3}{9} = \frac{21}{9} = \frac{7}{3}$

$0.2\bar{3} = \frac{23}{90}$

Content

- Classification of numbers.
- Divisibility rules.
- Factorials.
- Unit digit calculation.

Types of numbers



Conversion of a decimal number to fraction:

Example:

$3.\overline{713} =$

Solution:

$$3.\overline{713} = 3 + \frac{713}{999} = \frac{2997 + 713}{999} = \frac{3710}{999}$$

Example:

$12.3\overline{45}=$

Solution:

Here only 45 are recurring.

Therefore, $12.3\overline{45} = 12 + \frac{345 - 3}{990} = 12 + \frac{342}{990} = 12 + \frac{38}{110} = 12 + \frac{19}{55} = \frac{679}{55}$

Question. Find the rational form of the recurring rational 0.5333333333

1/2

[A] 11/99

[B] 1/3

[C] 8/15

[D] 13/30

Question. Convert 37.565656565656................ into P/Q

A)3719/99

B)3663/90

C)3443/990

D)None of these

$37.\overline{56}$

$37 + 0.\overline{56}$

$37 + \frac{56}{99}$

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UNIT DIGIT

218
↑
unit Digit

$12 \times 13 + 4 = 160$

$12 \times 11 - 2 = 130$

$13 \times 10 + 3 = 133$

$133 \times 52 + 691 - 332 \times 998$

$= 6 + 1 - 6 = 1$

Unit of Exponents

0 to 9

- → 0, 1, 5, 6
- → 4, 9
- → 2, 3, 7, 8

$0, 1, 5, 6 \rightarrow$ Any Power $\rightarrow$ No. itself in the unit place.

$10^1 = 10, 10^2 = 100, 10^3 = 1000$

$1^1 = 1, 1^2 = 1, 1^3 = 1, \ldots$

$5^1 = 5, 5^2 = 25, 5^3 = 125, 5^4 = 625 \ldots$

$6^1 = 6, 6^2 = 36, 6^3 = 216 \ldots$

$(312585)^{312585}$

$= 5$

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$\boxed{4, 9}$

$4^1 = 4$, $4^2 = 6$, $4^3 = 4$, $4^4 = 6$

$9^1 = 9$, $9^2 = 1$, $9^3 = 9$, $9^4 = 1$

$4^{odd} = 4$, $4^{even} = 6$

$9^{odd} = 9$, $9^{even} = 1$

$\boxed{2, 3, 7 \ \& \ 8}$

$2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 6$

$2^5 = 2$, $2^6 = 4$, $2^7 = 8$, $2^8 = 6$

$2^9 = 2$, $2^{10} = 4$

2^{41}

$\frac{41}{4}$, Rem = 1

$2^1 = 2$

Q

$\cancel{(3)}(3)^{49}$

$(3)^{49}$

$\frac{49}{4}$, Rem = 1

$(3)^1 = 3$

$(7)^{99}$

$\frac{99}{4} = R = 3 \quad (7)^3$

$= 3$

$(8)^{102}$

$\frac{102}{4},\ R = 2 \quad (8)^2$

$= 4$

$$(2)^{80} = ?$$

1	2	3	4th
2	4	6	8
5th	6th	7th	8th
9th			12th

$$(2)^8 \qquad \frac{8}{4} \quad \boxed{R=0} \qquad 2^4 = 6$$

Rem=0, Treat it as 4

$(2)^{80}$

$\frac{80}{4}$, R=0

$2^4 = 6$

$(7)^{80} \longrightarrow 7^4 = 9 \times 9$

$= 1$ Ans

Question: Find the unit digit of $(2354)^{1048}$

[A] 4

[B] 6

[C] 8

[D] None of these

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Question: Find the unit digit of $(248)^{1587}$

[A] 2

[B] 4

[C] 8

[D] 6

$8^3 = 512$

$= 2$

Question: Find the unit digit of $(127)^{223}$

[A] 7

[B] 9

[C] 3

[D] 1

$7^3 = 343$

$= 3$

Question: Find the unit digit of $(456)^{87} \times (307)^{42}$

[A] 1

[B] 4

[C] 5

[D] 7

6×7^2

$6 \times 9 = 4$

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$$(2156)^{699} + (624)^{624} \times (699)^{99} - (233)^{233}$$

$$6 + \underline{6 \times 9} - 3'$$

$$6 + 4 - 3 = 7$$

special case of 5

$5 \times 1 = 5$

$5 \times 2 = 10$

$5 \times 3 = 15$

$5 \times 4 = 20$

$5 \times 5 = 25$

⋮

$\boxed{\begin{array}{l} 5 \times \text{odd} = 5 \\ 5 \times \text{even} = 0 \end{array}}$

① $1 \times 3 \times 5 \times 7 \times 9 \text{ — — — } \times 99$

$= 5$

② $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9$

$= 0$

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Q $\underbrace{1! + 2! + 3! + 4!}_{1+2+6+24} + 5! + \ldots 99!$

$0 + \ldots$

$1+2+6+24 \rightarrow 33 = 3$

Q $(2178)^{2178!}$

Q $1! \times 2! \times 3! \times 4! \ldots \times 99!$

$= 0$

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$$(2178)^{2178!}$$

$$\frac{2178 \times \_\_\_\_\_\_\_ \times \cancel{4} \times 3 \times 2}{\cancel{4}}$$

$$\boxed{R = 0}$$

$$(8)^4 = 8^2 \times 8^2$$

$$= 4 \times 4$$

$$= 6$$

$$\frac{32}{4} \qquad \frac{8 \times \cancel{4}}{\cancel{4}}$$

Div Rules

$2^n \Rightarrow 2, 4, 8, 16$ ----

$2 = 2^1$
$4 = 2^2$
$8 = 2^3$
$16 = 2^4$

check last n digits.

61 29

2189

by 4 or not

2778

16 or not

$5^n \Rightarrow 5, 25, 125 \rightarrow 0$

$5 = 5^1 \rightarrow 00$

$25 = 5^2$

$125 = 5^3 \rightarrow 000$

check last n digits

125	25		5
✗	✗	2158	✗
✗	✓	2100	✓
✗	✗	2385	✓

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$\boxed{3^{rd}}$ → 3, 9, 27 ― ― ― .

Sum of Digits must be divisible

$\underbrace{213987}_{30}$ ✗ ✓

$\boxed{11}$

sum(O) ~ sum(E)

Either zero or div by 11.

8

1 2 3 4 3 2 1   0

8

Variation

$6 \rightarrow 2 \times 3$

218923

$12 = \cancel{6 \times 2}$

$\boxed{4 \times 3}$

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Divisibility Rules

Divisibility by Number	Divisibility Rule
Divisibility by 2	The last digit should be even.
Divisibility by 3	The sum of the digits should be divisible by 3.
Divisibility by 4	The last two digits should be divisible by 4.
Divisibility by 5	The last digit should either be 0 or 5.
Divisibility by 6	The number should be divisible by both 2 and 3.

Divisibility by 8	The last three digits should be divisible by 8.
Divisibility by 9	The sum of the digits should be divisible by 9.
Divisibility by 10	The last digit should be 0.
Divisibility by 11	The difference of the alternating sum of digits should be divisible by 11.

Question: If number 259292N is divisible by 2. How many values N can take?

0, 2, 4, 6, 8

[A] 4

[B] 5

[C] 3

[D] 6

Question: What should come in place of N if 545K5 is divisible by 9?

[A] 7

[B] 8

[C] 9

[D] 2

19+K

8

Question: For what values of P number 36345472P34 is exactly divisible by 9.

[A] 3

[B] 4

[C] 6

[D] 7

$41 + P = 45$

$P = 4$

Question: For what values of N number 549724N is exactly divisible by 6.

[A] 2 & 8

[B] 4 & 6

[C] 2 & 6

[D] 6 & 8

31 + N

0, 2, 4, 6, 8

Question: For what values of K number 99857K32 is exactly divisible by 11.

27 16+K

11 16 K=0

[A] 1

[B] 0

[C] 3

[D] 4

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Question: What should come in place K if 4857K is divisible by 88?

11×8

1K

48576

[A] 6

[B] 8

[C] 2

[D] 4

[E] None of these

Factorials

- → exponents
- → Trailing zeroes

$n! = n \times n-1 \times n-2 \times n-3 \ldots$

$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$

Max power of Any Prime Base in a Factorial

Ques $\boxed{3^{max}}$ in $10!$

$10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

3×3

3×2

$= 3^4$

4 Ans

3^{MAX} in $20!$

3 6 9 12 15 18

Method 2

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

1 1 2 1 1 2

$\boxed{8}$

Quotient

$$\frac{20}{3} = \frac{6}{3} = 2 \Rightarrow \boxed{8}$$

Ques 7^{max} in $200!$

$$\frac{200}{7} = \boxed{\frac{28}{7} = 4} = \boxed{32}$$

Ques 5^{max} in $100!$

$$\frac{100}{5} = \frac{20}{5} = 4 = \underline{24}$$

MAX POWER OF COMPOSITE NO.

6^{max} in $30!$ $\boxed{14}$

$6 = 2 \times 3$

$\frac{30}{2} = \boxed{\frac{15}{2} = \frac{7}{2} = \frac{3}{2} = 1} = 26$

$\frac{30}{3} = \boxed{\frac{10}{3} = \frac{3}{3} = 1} = 14$

$2^{26} \times 3^{14} \Rightarrow 2^{14} \times 2^{12} \times 3^{14}$

$= (6)^{14}$

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Lower Power → Ans

↳ Higher No.

6^{MAX} $\quad 30!$

↳ 3×2

$$\frac{30}{3} = \frac{10}{3} = \frac{3}{3} = 1 = 14$$

Q 21^{MAX} in $200!$

7×3

$\frac{200}{7} = \frac{28}{7} = 4$

32

Q 15^{MAX} in $150!$

5×3

$\frac{150}{5} = \frac{30}{5} = \frac{6}{5} = 1$

$= 37$

Trailing Zeroes

30 → 1

3030 → 1

10 → 1 zero → 10^1

100 → 2 zeroes → 10^2

1000 → 3 zeroes → 10^3

Trailling zeroes in 100!

Max power of 10

5 2

$$\frac{100}{5} = \frac{20}{5} = 4 = 24$$

No. of trailing zeroes $= 10^{MAX}$

$= 5^{MAX}$

Factorials

Trailing zeros (Number of zeros at the end)

Concept: In multiplication zero can be produced when we have a pair of 2 and 5. So, in order to find no. of zeros at the end, the pair of 2 and 5 need to be counted.

Example: $2^3 \times 5 \times 9 \times 8 \times 24 \times 15$ Find number of zeros at the end.

Question: Find number of trailing zeros in 50!

[A] 12

[B] 10

[C] 15

[D] 27

$\frac{50}{5} = \frac{10}{5} = 2 = 12$

Question: Find number of trailing zeros in 154!

[A] 35

[B] 31

[C] 34

[D] 37

$\frac{154}{5} = \frac{30}{5} = \frac{6}{5} = 1 = \boxed{37}$

Question: Find number of trailing zero in 100!+200! +300!

[A] 24

[B] 49

[C] 73

[D] None of these

$\frac{100}{5} = \frac{20}{5} = 4 = 24$

(32) 1000

(22) 100

(12) 10

1110

1 zero

Question: Find number of trailing zero in 100! X 200!

[A] 24

[B] 49

[C] 73

[D] None of these

24 | 49

$\frac{200}{5} = \frac{40}{5} = \frac{8}{5} = 1$

$10^{24} \times 10^{49} \Rightarrow 10^{73}$

Factors

1. Total No. of factors
2. Total No. of Prime factors
3. Total No. of Unique Prime factors
4. TN of odd factors
5. TN of Even factors
6. Product of all factors
7. Sum of all factors
8. factors which are perfect squares
9. Factors which are Perfect Cubes

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Factors = Divisor

① $\boxed{\text{Total No. of factors}}$

72 → 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72

12 Factors

$2^0 3^0 \to 1$	$2^2 3^0 \to 4$
$2^0 3^1 \to 3$	$2^2 3^1 \to 12$
$2^0 3^2 \to 9$	$2^2 3^2 \to 36$
$2^1 3^0 \to 2$	$2^3 3^0 \to 8$
$2^2 3^0 \to 4$	$2^3 3^1 \to 24$
$2^3 3^0 \to 8$	$2^3 3^2 \to 72$

$72 \to 8 \times 9$

$\to 2^3 \times 3^2$

$4 \times 3 = \boxed{12}$

$2^0, 2^1, 2^2, 2^3$

$3^0, 3^1, 3^2$

$N = a^p b^q c^r$

$\boxed{TNF = (p+1)(q+1)(r+1)}$

$\boxed{1800} \rightarrow 18 \times 100$

$2 \times 9 \times 4 \times 25$

$\rightarrow 2^3 \times 3^2 \times 5^2$

$4 \times 3 \times 3 = \boxed{36}$

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② TN of Prime Factors → Sum of all Powers

③ TN of Unique Prime factors

↳ Total Prime Bases

1800 → $2^3 \times 3^2 \times 5^2$

② TN of Prime Factors = 3+2+2

= 7

③ TN of Unique Prime Factors

$= \boxed{3}$

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(4) odd Factors → Ignore Power of 2.

(5) Even Factors

$1800 \rightarrow 2^3 \times 3^2 \times 5^2$

$\Downarrow$

$3^2 \times 5^2$

$TNF = 3 \times 3 = 9$

Total − odd = Even.

$36 - 9 = 27$

$1800 \rightarrow 2^3 \times 3^2 \times 5^2$

$\cancel{2^0}$ odd	3^0	5^0
2^1 even	3^1	5^1
2^2 even	3^2	5^2
2^3 even		
3	3	3

$= 27$

5 → Product of Factors

7 → Sum of Factors

6

72 → 1 2 3 4 6 8 9 12 18 24 36 72

$\boxed{TF = 12}$

$(72)^6 =$

$$\boxed{Prod = (N)^{\frac{TNF}{2}}}$$

Sum

$$72 = 2^3 \times 3^2$$

$2^0 \quad 3^0$

$2^1 \quad 3^1$

$2^2 \quad 3^2$

2^3

$$2^0 3^0 + 2^0 3^1 + 2^0 3^2 + 2^1 3^0 + 2^1 3^1 + 2^1 3^2 + 2^2 3^0 + 2^2 3^1 + 2^2 3^2 + 2^3 3^0 + 2^3 3^1 + 2^3 3^2$$

$$2^0(3^0+3^1+3^2) + 2^1(3^0+3^1+3^2) + 2^2(3^0+3^1+3^2) + 2^3(3^0+3^1+3^2)$$

$$\boxed{= (2^0+2^1+2^2+2^3)(3^0+3^1+3^2)}$$

$$\text{Sum} = N = a^p b^q c^r$$

$$\boxed{(a^0 + a^1 + \cdots a^p)(b^0 + b^1 + \cdots b^q)(c^0 + c^1 \cdots c^r)}$$

$$1800 = 2^3 \times 3^2 \times 5^2$$

$$(2^0 + 2^1 + 2^2 + 2^3)(3^0 + 3^1 + 3^2)(5^0 + 5^1 + 5^2)$$

$$= \text{Ans}$$

(8) Factors which are Perfect Squares

(9) Factor which are Perfect Cubes

$1800 \rightarrow 2^3 \times 3^2 \times 5^2$

2^0 ✓ , $\cancel{2^1}$, 2^2, $\cancel{2^3}$

3^0 ✓ , $\cancel{3^1}$, 3^2 ✓

5^0 ✓ , $\cancel{5^1}$, 5^2 ✓

$\overset{PS}{1} \times \overset{PS}{4} = \overset{PS}{4}$

$\overset{PS}{4} \times \overset{PS}{9} = \overset{PS}{36}$

$\overset{PS}{9} \times \overset{PS}{16} = \overset{PS}{144}$

$2 \times 2 \times 2 = \boxed{8}$

$$1800 \rightarrow 2^3 \times 3^2 \times 5^2$$

$$\begin{array}{ccc} 2^0 & 3^0 & 5^0 \\ 2^1 & 3^1 & 5^1 \\ 2^2 & 3^2 & 5^2 \\ 2^3 & & \end{array}$$

$$2 \times 1 \times 1 = \boxed{2}$$

Factors which are Perfect Squares

$1800 \rightarrow 18 \times 100 \rightarrow 9 \times 2 \times 4 \times 25$

$2^3 \times 3^2 \times 5^2$

$1 \times 4 \rightarrow 4$

$4 \times 9 = 36$

$9 \times 16 = 144$

$2^0 \quad 3^0 \quad 5^0$

$2^1 \quad 3^1 \quad 5^1$

$2^2 \quad 3^2 \quad 5^2$

2^3

$2 \times 2 \times 2$

$= \boxed{8}$

$2^3 \times 3^2 \times 5^2$

$2^0 \quad 3^0 \quad 5^0$

$2^1 \quad 3^1 \quad 5^1$

$2^2 \quad 3^2 \quad 5^2$

2^3

$2 \times 1 \times 1 = 2$

1. TNF ✓ $5 \times 4 \times 4 = \boxed{80}$
2. TN of Prime $4+3+3 = \boxed{10}$
3. TN of Unique Prime $= \boxed{3}$
4. odd $\rightarrow 3^3 \times 5^3 \Rightarrow 4 \times 4 = \boxed{16}$
5. Even $\rightarrow 80-16 = \boxed{64}$
6. Prod $\rightarrow (54000)^{40}$
7. Sum $(2^0+2^1+2^2+2^3+2^4)(3^0+3^1+3^2+3^3)(5^0+5^1+5^2+5^3)$
8. perfect Sq $\rightarrow 3 \times 2 \times 2 = \boxed{12}$
9. ~~Perfect cube~~ $2 \times 2 \times 2 = \boxed{8}$

$54000 = 54 \times 1000 \rightarrow 2 \times 27 \times 2 \times 125$

$\rightarrow \underline{2}^4 \times \underline{3}^3 \times \underline{5}^3$

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LCM & HCF

Least common $\boxed{\text{multiple}}$

divisible by

Limited

Highest Common $\boxed{\text{Factor}}$

divides

Unlimited

1, 2, 3, 6

6

6, 12, 18, 24 ......

Factors

Multiples

12, 18

HCF

→ Highest Common Factor

12 → 1, 2, 3, 4, 6, 12

18 → 1, 2, 3, 6, 9, 18

HCF = 6

LCM → Least Common Multiple

12 → 12, 24, 36, 48, 60, 72 — — - -

18 → 18, 36, 54, 72, 90 — — - -

LCM = 36

12, 18

HCF = 6

LCM → 36

Prime Factorise

12 → $2^2 \times 3$

18 → 2×3^2

$6 = 2 \times 3$

$36 = 2^2 \times 3^2$

HCF → minimum Power of the Prime Base.

LCM → maximum Power of the Prime Base.

$N_1 = 2^2 \times 3^2 \times 5 \times 7 \times 11^0$

$N_2 = 2^3 \times 3 \times 5^2 \times 7 \times 11$

$HCF = 2^2 \times 3 \times 5 \times 7$

$LCM = 2^3 \times 3^2 \times 5^2 \times 7 \times 11$

Division method

HCF

$$\begin{array}{r|l} 2 & 12, 18 \\ \hline 3 & 6, 9 \\ \hline & 2, 3 \end{array}$$

$HCF = 3 \times 2 = 6$

$$\begin{array}{r|l} 2 & 12, 18 \\ \hline 3 & 6, 9 \\ \hline 2 & 2, 3 \\ \hline 3 & 1, 3 \\ \hline & 1, 1 \end{array}$$

$Lcm = 2 \times 3 \times 2 \times 3$

$= 36$

18, 12

216

HCF = 6, LCM = 36

216

$$\boxed{N_1 \times N_2 = LCM \times HCF}$$

HCF is always Factor of LCM.

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12, 18

$$\begin{array}{l|l} 2 & 12, 18 \\ \hline 3 & 6, 9 \\ \hline & 2, 3 \end{array}$$

Ratio of the No.s

HCF = 2 x 3

$LCM = \boxed{2 \times 3} \times 2 \times 3$

LCM = HCF x Product of Ratios

12, 18 HCF = 6, LCM = 36

2 : 3

12 18

Ratio x HCF = No.

$$\text{LCM of Fractions} = \frac{\text{LCM (Nr)}}{\text{HCF (Dr)}}$$

$$\text{HCF of fractions} = \frac{\text{HCF (Nr)}}{\text{LCM (Dr)}}$$

LCM & HCF

LCM- Least Common Multiple

For e.g. 12= 12, 24, 36, 48, 60, 72, 84, 96............

16= 16, 32, 48, 64, 80, 96............

So there are lots of multiple of 12 and 16 which will be common but the least multiple which is common is 48.

So LCM = 48

HCF- Highest Common Factor

For e.g. 12= 1, 2, 3, 4, 6, 12

16= 1, 2, 4, 8, 16

So there three common factor=1, 2, 4 but the Greatest factor is 4. That is the HCF of 12 and 16.

Important Formulae

Product of Two numbers = HCF x LCM

$$\text{HCF of Fraction} = \frac{\text{HCF of numerators}}{\text{LCM of Denominator}}$$

$$\text{LCM of Fraction} = \frac{\text{LCM of numerators}}{\text{HCF of Denominator}}$$

LCM Word Problems

Question	Answer
The Least number which is exactly divisible by x, y and z	LCM(x, y, z)
The least number which when divided by x, y, z leaves the same remainder R in each case	LCM (x, y, z) + R
The least Number which when divided by x, y and z, leaves remainder a, b and c respectively	LCM (x, y, z)–K Where K=(x-a)=(y-b)=(z-c)

HCF Word Problems

Question	Answer
The greatest number that will exactly divide x, y and z	HCF(x, y, z)
The greatest number that will divide x, y, z and leaves remainder R in each case	HCF(x−y, y−z, z−x) OR HCF(x−R, y−R, z−R)
The greatest Number that will divide x, y and z, leaving remainder a, b and c respectively	HCF(x−a, y−b, z−c)

Q) Find the lowest common multiple of 24, 36 and 40.

[A] 120

[B] 240

[C] 360

[D] 480

$24 = 2^3 \times 3$

$36 = 2^2 \times 3^2$

$40 = 2^3 \times 5$

$2^3 \times 3^2 \times 5$

$8 \times 9 \times 5$

$40 \times 9 = 366$

Q) The greatest number that will exactly divide 36 and 84 is:

[A] 4

[B] 6

[C] 12

[D] 18

Q) Four bells ring at an interval 3min, 4min, 5min and 6 minutes respectively. If all the four bells ring at 9am first, when will it ring again?

A]. 10:00am

B]. 10:30am

C]. 11:00am

D]. None of these

9AM

3 4 5 6

6 8 10 12

LCM 60

9:AM + 60 min = 10:00 AM

Q) Which of the following fraction is the largest? 7/8, 13/16, 31/40, 63/80

[A] 7/8

[B] 13/16

[C] 31/40

[D] 63/80

7/8 ×80 10 → 70

13/16 ×80 5 → 65

31/40 ×80 2 → 62

63/80 ×80 → 63

Q) Three number are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:

[A] 40

[B] 80

[C] 120

[D] 200

$LCM = HCF \times 3 \times 4 \times 5$

$\cancel{2400} = HCF \times \cancel{3 \times 4 \times 5}$

120

40

$HCF = 40$

Q) Find the smallest number, which when divided by 3, 4 and 5 leaves remainder 1, 2 and 3 respectively?

[A] 60

[B] 53

[C] 58

[D] None

Q) The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively, is:

[A] 123

[B] 127

[C] 235

[D] 305

Q) Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

[A] 4

[B] 7

[C] 9

[D] 13

Q) The product of two numbers is 4107. If the H.C.F. of these numbers is 37, then the greater number is:

[A] 101

[B] 107

[C] 111

[D] 185

Remainders

$$\begin{array}{r|l} 5 & 32 \; | \; 6 \\ & \underline{30} \\ & \;\; 2 \end{array}$$

6 ← Quotient

32 → Dividend

2 → Remainder

5 → Divisor

$$32 = 5 \times 6 + 2$$

$$\boxed{\text{Dividend} = \text{Div} \times Q + R}$$

Question: In a division process divisor is 5 times of quotient and remainder is 3 times of quotient. If remainder is 15. Find dividend.

[A] 140

[B] 150

[C] 170

[D] 190

R

$3Q = 15$

$\boxed{Q=5}$

$D = 5 \times 5 = 25$

$N = 25 \times 5 + 15$

$= 125 + 15 = 140$

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$$\frac{\overset{2}{12} \times \overset{3}{13}}{10} = \frac{156}{10} \quad Rem = 6$$

Rem

$$2 \times 3 = 6$$

* Divisor can be used multiple times.
* Rem < Divisor

$$
\begin{array}{r|l}
5 & 326 \\
 & 30 \\
\hline
 & 2
\end{array}
\qquad
\begin{array}{r|l}
5 & 327 \\
 & 35 \\
\hline
 & -3
\end{array}
$$

Positive − Negative = Divisor

$2-(-3) = 5$

$\boxed{5}$

$\boxed{2}$

$2-5 = -3$

5

−3

$5-3 = 2$

Question: A number when divided by 252 leaves the remainder 34. Find the remainder when the same number is divided by 21.

$N = 252 \times Q + 34$

21

13

[A] 13

[B] 5

[C] 7

[D] 2

Cannot

1 2

1 2

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Important Rules

1. If x/D Remainder= R, then 2x/D Remainder= 2R/D
2. If x/D Remainder= R, then $\frac{x^2}{D}$ Remainder= R^2/D
3. If x/D Remainder= R, then $\frac{x^3}{D}$ Remainder= R^3/D
4. If $\frac{A \times B}{D}$ then Remainder= $\frac{Ar \times Br}{D}$ (product of individual remainders)
5. If $\frac{A+B}{D}$ then Remainder= $\frac{Ar+Br}{D}$ (Sum of individual remainders)

Question: Find remainder for $\frac{63 \times 78}{9}$ → 0

Question: Find remainder for $\frac{63+78}{9}$

0+6 = 6

Question: When a number divided by a divisor m then remainder is 20. If the twice of that number is divided by the same divisor m then the remainder is 9. Find the divisor m.

8

$$\frac{18^{20}}{17}$$ Rem = ?

$$\frac{18 \times 18 \times 18 \times \text{------ 20times}}{17}$$

$$= (1)^{20} \Rightarrow 1$$

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$$\frac{17^{20}}{18} \quad \text{Rem} = ?$$

$$(-1)^{20} \rightarrow 1$$

$$\frac{19^{21}}{20} \quad \text{Rem} = ?$$

$$\hookrightarrow (-1)^{21} = -1$$

$$20 - 1 = 19$$

Important Formulas:

1. $\frac{(x+a)^n}{x}$ then remainder= a^n

2. $\frac{(tx+a)^n}{x}$ then remainder= a^n

3. $\frac{(x+1)^n}{x}$ then remainder= $1^n = 1$

4. $\frac{(x-a)^n}{x}$ then remainder= $(-1)^n$ if power even then R=1
   if Power Odd then R=-1 or x-1

5. $\frac{(tx+1)^n}{x}$ then remainder= $1^n = 1$

6. $\frac{(tx-1)^n}{x}$ then remainder= $(-1)^n$ if power even then R=1
   if Power Odd then R=-1 or x-1

Remainder Cyclicity:

Example: What is the remainder when 2^201 is divided by 7?

2^1/7 = R(2)

2^2/7= R(4)

2^3/7 = R(1) or 8/7 = R(1)

Therefore [(2^3)^67] / 7 = 1 / 7

1/7 = R(1)

Note: Do not cancel any numerator value with the denominator value as the remainder will differ.

$R(6/4) \neq R(3/2)$

$6/4 = R(2)$

But $3/2 = R(1)$

If you want to cancel out numerator and denominator by **a certain value** then at last we also need to multiply the remainder by the **same value** in order to get correct remainder

Q) What is the remainder when 2^7 is divided by 8?

[A] 0

[B] 4

[C] 5

[D] 7

0

$$\frac{2^3 \times 2^4}{8} = \boxed{0}$$

Q) Remainder when 35^{113} is divided by 9?

[A] 1

[B] 8

[C] 3

[D] 4

$(-1)^{113} \rightarrow -1$

$9 - 1 = 8$

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Q) Remainder when 2^{45} is divided by 9?

[A] 1

[B] 4

[C] 8

[D] 5

$$\frac{(2^3)^{15}}{9}$$

$$\frac{(8)^{15}}{9}$$

$$(-1)^{15} = -1$$

$$9 - 1 = 8$$

Q) Remainder when 2^{99} is divided by 10?

[A] 1

[B] 4

[C] 2

[D] 8

$\text{Rem}\ \frac{N}{10}$ = Unit Digit of N

$\text{Rem}\ \frac{N}{100}$ = Last 2 Digit of N

$2^{99} \rightarrow \frac{99}{4} = \boxed{3}$

$2^3 = \boxed{8}$ Ans

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Multiplying Factor

$\frac{\cancel{38}}{\cancel{4}}$ Rem = 2

$\boxed{2}$

$\frac{19}{2} \Rightarrow 1$

$2 \times 1 = \boxed{2}$

$$\frac{2^{96}}{96} \qquad \text{Rem} = ?$$

$$\boxed{2^5}$$

$$\frac{2^{96}}{32 \times 3} \Rightarrow \frac{2^{96}}{2^5 \times 3} \Rightarrow \boxed{\frac{2^{91}}{3}}$$

$$(-1)^{91} = -1$$

$$3 - 1 = \boxed{2}$$

$$2 \times 2^5$$

$$2 \times 32 = 64$$

$$\frac{3^{3^2}}{36} \qquad \text{Rem} = ?$$

$$\frac{3^{3^2}}{9 \times 4} \Rightarrow \frac{3^{\cancel{3^2}}}{\cancel{3^2} \times 4}$$

$$\frac{3^{30}}{4} = (-1)^{30}$$

$$= 1$$

$\boxed{3^2}$

$$1 \times 3^2 = 9$$